

ARTICLE

Multiple Approaches to Improve Multi-Label Emotion Classification in Korean Texts Using Semi-Supervised Korean BERT

Jisuk Han¹, Juhyang Kim¹, Dayoung Shim¹, Alfin Whilliam Adi Putra¹, Xiaopeng Yang¹ and Jae Do Yang^{2,3,4}

¹School of Applied AI and Entrepreneurship, Handong Global University, Pohang, Korea, ²Department of Surgery, Jeonbuk National University Medical School and Hospital, Jeonju, Korea, ³Research Institute of Clinical Medicine, Jeonbuk National University, Jeonju, Korea, and ⁴Biomedical Research Institute, Jeonbuk National University Hospital, Jeonju, Korea

Corresponding authors: Xiaopeng Yang; Email: ryp233@handong.edu; Jae Do Yang; Email: yjd@jbnu.ac.kr

(Received xx xxx xxx; revised xx xxx xxx; accepted xx xxx xxx)

Abstract

Emotions in the Korean language are expressed through subtle linguistic cues that require deep cultural and contextual understanding. Emotion analysis in Korean, therefore, represents a challenging area of natural language processing that extends beyond simple positive–negative sentiment classification to capture complex, co-occurring feelings. This study aims to develop a Korean-language emotion classification model using multi-label classification to capture co-occurring and nuanced emotional expressions. To enhance the model’s performance, we utilize a combination of real-world and synthetic data, including the Korean Online That-gul Emotions (KOTE) dataset, Naver blog posts, and ChatGPT-generated diary entries. We apply a multi-level labeling approach based on pseudo-labeling, in which the model is initially trained on labeled data and then iteratively refined using self-labeled, previously unlabeled data. This semi-supervised framework expands emotional coverage across 43 emotion categories and achieves robust performance across diverse text domains. By combining heterogeneous data and adaptive self-training methods, this research contributes to the development of scalable, reproducible, and ethically responsible emotion recognition systems for the Korean language.

Keywords: multi-label emotion classification; Korean text; Korean BERT; sentiment analysis

1. Introduction

Emotion is a complex psychological state that integrates physiological responses, cognitive appraisals, and expressive behaviors, and is a key component of human interaction and mental processes (Bakker and Daniels, 2012; Scherer et al. 2004). In the domain of human-computer interaction (HCI), accurate recognition of a user’s emotional state enables more personalized and empathetic digital systems, enabling a more intuitive user experience (Martínez-Tejada et al. 2020). Traditional sentiment analysis typically categorizes texts into three categories: positive, negative, and neutral, but recent research has shown that this approach has limitations in capturing the subtle and varied emotional expressions reflected in natural language (Acheampong et al. 2020).

To compensate for these limitations, this study proposes a sophisticated emotion classification framework based on Korean diary data. Diaries are ‘real-time’ records of an individual’s inner, subjective emotional experiences and are a source of qualitative data that allows for in-depth exploration of emotional and behavioral patterns embedded in them (Bolger et al. 2003; Nadin and Cassell, 2006). Existing research also suggests that journaling promotes self-understanding and emotion regulation, which has a positive impact on an individual’s mental health and quality of life (Briner and Kiefer, 2005).

Furthermore, emotional experience is a culturally constructed concept, and its expression is not universal but rather depends on the socio-cultural context (Mesquita and Frijda, 1992; Gendron et al. 2014). As a result, languages, especially those of high-context cultures, tend to imply emotions in subtle and multi-layered ways, which leads to the presence of more complex and layered emotions within a single text or sentence (Cowen and Keltner, 2017; Strapparava and Mihalcea, 2007). Therefore, in this study, we utilize the Korean Online That-gul Emotions (KOTE) dataset (Jeon, Lee, and Kim, 2022), which contains 43 fine-grained emotion labels that reflect the emotional structure of Korean speakers. KOTE adopts a multi-label-based annotation scheme under the premise that a single sentence can express multiple emotions simultaneously. Considering these linguistic and cultural specificities, this study proposes a hybrid deep learning model that combines contextual Korean BERT (Bidirectional Encoder Representations from Transformers) variants with morpheme-level FastText embeddings. This dual embedding structure enables more accurate and interpretable sentiment classification by reflecting both sentence-wide context and word-level semantic information. In addition, we introduce a correlation-aware classification technique that reflects the correlation between emotions and then compare its classification performance among various emotion categories.

Ultimately, this study aims to explore and implement novel techniques to enhance the model's ability to capture the nuanced emotional expressions inherent in the Korean language. To achieve this, we adopt a comprehensive training approach that begins with a semi-supervised learning framework, incorporating multi-level pseudo-labeling and data augmentation to maximize the model's performance (Yu et al. 2021; Yang et al. 2023).

2. Related Work

2.1 Emotion Analysis

Emotion analysis is a technology that identifies and classifies emotions through human language expressions, behaviors, and physiological responses. It is a field of research that aims to extract and analyze nuanced emotions from linguistic input, enabling machines to interpret emotional content with greater sophistication (Acheampong et al. 2020). Emotion analysis has developed rapidly alongside advancements of information technology and the growing demand for human-centered interfaces. It has been widely used to understand users' emotional state and provide useful feedback based on that understanding, expanding into various fields.

Early emotion analysis began with simple sentiment analysis. Notable examples include Pang et al. (2002)'s research, which proposed a machine learning-based approach to classify the positivity or negativity of text using movie review data, and Turney (2002)'s research, which utilized unsupervised learning to introduce the concept of semantic orientation and automated sentiment analysis by classifying the emotional orientation of sentences.

Building on early studies that demonstrated the effectiveness of sentiment analysis, subsequent research sought to identify more sophisticated and multilayered emotional structures beyond simple binary classification, leading to the exploration of various emotion classification systems. In 2007, research on emotion classification in news headlines as part of the SemEval task demonstrated the need for diverse emotional labels and proposed a new approach to multi-emotion detection (Strapparava and Mihalcea, 2007). In 2020, Google's GoEmotions project introduced a 27-category emotion classification system, emphasizing the need for advanced analytical models capable of capturing the complexity and subtlety of human emotions (Demszky et al. 2020).

Recently, sentiment analysis has been applied to a wide range of fields, including public health surveillance, personalized education, and mental health diagnosis (Akar, Kim, and Noh, 2024).

2.2 Multi-label classification

In sentiment analysis, multi-label classification is a method designed to assign multiple sentiment labels to a single word or sentence simultaneously. It is an essential technique for reflecting the multilayered, complex, and simultaneous nature of emotions (Ortony, Clore, and Collins, 1990). Early sentiment analysis research focused on binary classification, which categorizes emotions in text as either positive or negative. This approach demonstrated a certain level of validity when applied to explicit emotion expression data, such as movie reviews (Pang et al. 2002; Turney, 2002). However, it has consistently been noted that human emotional expression in real-world contexts is far more complex and cannot easily be reduced to a single emotion (Strapparava and Mihalcea, 2007).

To address these limitations, multi-label emotion classification methods were developed to better capture the emotional complexity and mixed nature of users' language. The GoEmotions project subdivided emotion labels into 27 categories and adopted a multi-label framework in which each sentence may be assigned two or more labels. This approach is based on the observation that actual Reddit sentences are rarely explained by a single emotion. According to the study, it is common for a single sentence to simultaneously express emotions such as "joy" and "surprise" or "admiration" (Demszky et al. 2020).

This trend is also evident in Korean emotional expressions. In the KOTE dataset, an average of approximately 7.91 emotion labels were assigned to a single comment (Jeon, Lee, and Kim, 2022). This finding suggests that the context-dependent and overlapping expressive characteristics of the Korean language allow complex emotions to coexist within a single utterance, making it difficult to accurately represent users' emotional states using a single-label classification method.

The increase in the number of emotion categories reflects not only technical advancements but also sociocultural factors that have diversified the ways in which people experience and express emotions (Cowen and Keltner, 2017). These scholars argue that emotions cannot be reduced to six basic categories, including joy, anger, sadness, disgust, fear, and surprise. Instead, multiple emotion categories are organically interconnected within real-world data. In particular, emotions such as "admiration", "pride", "discomfort", and "inspiration" are not fully accounted for in traditional emotion theories, yet they frequently appear in modern social media and conversational texts (Jabreel and Moreno, 2019).

Because such complex emotions are often expressed simultaneously within a single text, multi-label classification is essential for realistic and sophisticated emotion analysis. This approach enables models to represent mixed emotional states and more accurately interpret the nuanced emotional dimensions of language (Revelle and Scherer, 2009).

2.3 Context of Korean culture and emotion

Recently, personality tests such as the MBTI (Myers-Briggs Type Indicator) have gained significant popularity in Korean society. These tests are used not merely for entertainment or curiosity, but also as tools for understanding oneself and others (Lee et al. 2023). Particularly among the MZ generation (Millennials and Generation Z), there is a growing trend toward seeking relational harmony through "knowing oneself" and "understanding others". This phenomenon is closely connected to the relationship-oriented and emotionally sensitive characteristics of Korean culture (Yoon and Kim, 2025). East Asian societies have traditionally been strongly collectivist, with cultural norms that emphasize sensitivity to others' perspectives and emotional states (Mesquita and Frijda, 1992; Revelle and Scherer, 2009). Consequently, the ability to understand and coordinate with others' tendencies and emotions is regarded as an important social competence.

Personality assessments such as the MBTI are classification tools based on relatively stable tendencies or response patterns, and they can contribute to understanding individuals to a certain extent. However, because these frameworks emphasize relatively enduring traits, they have limitations in capturing fluid emotional states in real time. While personality accounts for consistent behavioral patterns, emotions provide more concrete and

context-sensitive information, as they are psychological phenomena that fluctuate depending on situational factors and arise from immediate reactions to external stimuli (Scherer et al. 2004). Particularly, emotions are expressed instantaneously through language, facial expressions, and tone of voice, thereby serving as direct clues for understanding others during interaction (Gendron et al. 2014).

Furthermore, because personality assessments are primarily based on self-report surveys, discrepancies may arise between how individuals perceive themselves and how they are perceived by others. In contrast, emotions can be directly inferred by external observers through linguistic expressions and behavioral responses, enabling a more observable and data-driven form of analysis (Demszky et al. 2020).

If personality tests serve as tools for understanding a person's temperament, emotion analysis functions as a tool for understanding a person's state. Particularly, emotion analysis offers the advantage of tracking relational dynamics in real time, which is increasingly important in contemporary society, where interpersonal relationships evolve rapidly and digital communication is pervasive (Martínez-Tejada et al. 2020). Therefore, this study adopts emotion as the central analytical unit within the generational trend of self-understanding and understanding others in Korean society, aiming to examine more realistic and dynamic aspects of human interaction.

2.4 Cultural Semantics of Korean Emotion Expression

Korean is a high-context language characterized by a complex interplay between linguistic structure and cultural context, and it exhibits distinctive features in emotional expression. For example, subjects are frequently omitted, and speakers often convey emotions indirectly through implicit statement or metaphorical expressions rather than stating them explicitly (Akar, Kim, and Noh, 2024). As a result, the same sentence may be interpreted as expressing entirely different emotions depending on the context, the speaker's disposition, and the listener's social status. This characteristic highlights the difficulty of accurately analyzing Korean emotional expressions using keyword-based or rule-based approaches, as well as English-centric pre-trained models (Jeon, Lee, and Kim, 2022).

Furthermore, the Korean honorific system plays a crucial role in shaping emotional nuance. Emotional expressions vary according to the speaker's social position and relationship with the listener, thereby increasing the complexity of analysis (Yoon and Kim, 2025). For instance, the expression “나는 괜찮아” “I’m fine” may convey a range of emotional states, such as “난 진짜 괜찮아” “I’m actually fine”, “난 안 괜찮아 근데 뭐” “I’m not saying I’m fine”, “괜찮다고” “I’m saying I’m okay”, or “나는 진짜 안 괜찮아” “I’m actually angry” depending on the situational and relational context. Because a single sentence can encompass multiple emotional layers, Korean emotion analysis must prioritize contextual interpretation rather than rely solely on isolated lexical units. Given that emotions are often implicitly expressed, in-depth interpretation is required, taking into account not only sentence structure, but also conversational flow, interpersonal relationships, and broader situational context (Kim, Han, and Kim, 2024).

In addition to linguistic factors, cultural influences play a significant role in shaping emotional expression. According to Mesquita and Frijda (1992), emotions are culturally constructed experiences; therefore, their definitions, expressions, and evaluative standards differ across societies. Korean society is highly relationship-oriented and tends to regulate emotional expression carefully in consideration of others' reactions (Martínez-Tejada et al. 2020). This cultural tendency to imply rather than explicitly articulate emotions further complicates emotion categorization. As a result, sentiment analysis models must move beyond simple dichotomous classifications such as “happy” or “sad” and instead be capable of analyzing complex emotional configurations, such as identifying that a sentence simultaneously conveys joy, anxiety, and anticipation. This need has contributed to the development of more sophisticated analytical frameworks based on multi-label classification (Ameer et al. 2023).

Ultimately, sentiment analysis extends beyond purely linguistic challenges and requires cultural interpretation. Particularly for high-context languages such as Korean, analytical models must be

designed to comprehensively account for emotional expression patterns, social relationships, cultural norms, and related contextual factors.

3. Methodology

3.1 About BERT

BERT is a model that pre-trains deep language representations from unlabeled text by simultaneously considering bidirectional context (left and right context) at all layers. Thanks to this structure, it can be effectively fine-tuned for various natural language processing tasks, such as question answering and sentence inference, by simply adding an output layer without any complex architectural changes (Devlin et al. 2019), as shown in Figure 1.

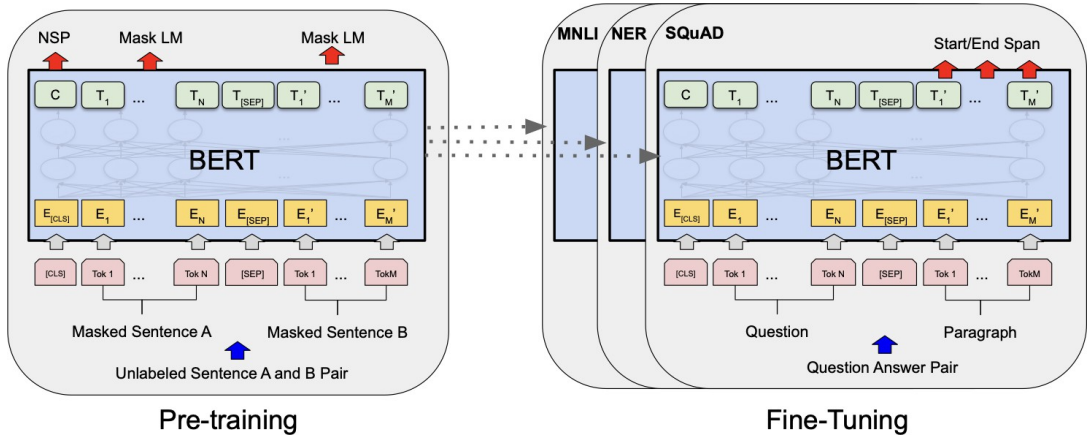


Figure 1. Pre-training and fine-tuning procedures for BERT (Devlin et al. 2019).

3.2 Types and characteristics of Korean BERTs

The original BERT model was pre-trained on a large-scale English corpus and, as a result, shows limited performance when applied to non-English languages such as Korean (Devlin et al. 2019). This limitation underscored the need for a bidirectional transformer model specifically tailored to Korean natural language processing (NLP) tasks. In response, several Korean-specific BERT models were developed. Although Google's multilingual BERT was designed to process multiple languages simultaneously (Devlin et al. 2019), it generally exhibits lower representational capacity for individual languages, which can lead to reduced performance in Korean-related tasks. To address these limitations, various BERT variants optimized specifically for Korean have subsequently been proposed (Lee et al. 2020).

The Korean language features syllable-level composition and a rich system of inflectional morphology, necessitating distinct approaches to morphological analysis and word segmentation compared to many other languages. In particular, the presence of more than 10,000 orthographically distinct syllables, along with complex inflectional patterns, has prompted the development of specialized tokenization methods capable of effectively segmenting Korean text. These tokenization strategies play a critical role in improving the performance of Korean NLP systems (Lee et al. 2020). In this study, we examine several Korean-specific BERT-based models to compare their performance on Korean NLP tasks.

3.2.1 KoBERT

KoBERT (SKTBrain, 2019) was developed by SKTBrain using a large-scale Korean corpus that includes Korean Wikipedia and news articles. The model employs a SentencePiece-based tokenizer, which offers advantages such as simplified training, efficient deployment, and stable

initialization (Yang et al. 2023). KoBERT is primarily applied to natural language processing tasks, including sentence classification and sentiment analysis.

3.2.2 KoELECTRA

KoELECTRA (Park, 2020) is another model that utilizes the SentencePiece tokenization method. It was pre-trained on a large-scale Korean corpus and is based on the ELECTRA architecture, which is recognized for its efficient training and strong overall performance. KoELECTRA has demonstrated particularly strong results in emotion-related tasks, including sentiment analysis (Jang et al. 2022).

3.2.3 KcBERT

KcBERT (Lee, 2020) is a model specifically optimized for processing informal text, including internet slang, abbreviations, and profanities commonly found in online communities. Trained on data collected from Korean online forums and news comments sections, the model demonstrates strong performance in tasks such as comment sentiment analysis, hate speech detection, and social media monitoring (Yang, 2021).

3.3 How to analyze Korean sentences in 43 labels

The KOTE dataset consists of 50,000 online comments annotated with 43 distinct emotion labels. The comments were collected from 12 different platforms, including news websites, online communities, social media, e-commerce platforms, movie review sites, microblogs, and discussion forums. From a total pool of 3.2 million comments, 50,000 were sampled in a balanced manner across platforms to ensure diversity and representativeness. The annotation process involved 3,084 native Korean speakers, each of whom was allowed to assign up to 10 emotion labels per comment. To ensure high reliability, the labeling was refined through a three-step review process (Jeon, Lee, and Kim, 2022). The 43 emotion labels in the KOTE dataset are as follows: “fed up”, “shock”, “disgust”, “contempt”, “anger”, “irritation”, “dissatisfaction”, “preposterous”, “pathetic”, “distrust”, “disappointment”, “embarrassment”, “shame”, “guilt”, “gossip”, “fear”, “anxiety”, “compassion”, “sadness”, “sorrow”, “despair”, “exhaustion”, “laziness”, “reluctant”, “boredom”, “arrogance”, “resolute”, “NO EMOTION”, “realization”, “surprise”, “respect”, “interest”, “expectancy”, “welcome”, “care”, “attracted”, “excitement”, “joy”, “happiness”, “admiration”, “pride”, “gratitude”, “relief”, and “comfort”.

3.4 Tokenizer

3.4.1 KoELECTRA

The segmentation method used in this study relies on the basic tokenizer library provided in KoELECTRA, which follows ELECTRA’s method to tokenize a sentence and then determine whether the tokenization is correct or not. For example, when tokenizing the sentence “이 밥은 대박 맛있다” “The food is super delicious”, it is segmented into [이, 밥, ##은, 대, ##박, 맛, ##있다]. During pre-training, the Generator predicts which word should appear in a masked position. For instance, given a sentence such as “I [MASK] ate rice,” the Generator may infer candidate words like “delicious,” “warm,” or “freshly cooked,” assigning probabilities to each. The Discriminator then evaluates the entire sentence and determines whether each token is original (True) or replaced (False). This approach differs from the traditional masked language modeling used in BERT. In BERT, selected tokens are masked and the model learns by predicting the original words. In contrast, ELECTRA trains the Discriminator to identify replaced tokens rather than to recover masked ones. As a result, the model learns from more informative training signals and achieves improved training efficiency and performance (Clark et al. 2020).

3.4.2 KcBERT

KcBERT also employs the WordPiece tokenizer, as used in KoELECTRA. By training on unstructured text collected from online communities (e.g., comments and posts), it is specifically designed to handle realistic language phenomena such as colloquialisms, neologisms, typos, and spacing errors. For example, the sentence “이 밥은 대박 맛있ㄷㅋㅋ” can be segmented by the WordPiece tokenizer into [이, 밥, ##은, 대, ##박, 맛, ##있다, ㅋㅋ]. This approach effectively preserves meaningful subword units without requiring explicit morphological analysis. KcBERT is based on BERT architecture and is trained using the masked language modeling objective, in which certain tokens in the input sentence are masked and the model learns to predict them. This enables the model to capture bidirectional contextual information, making it well suited for downstream tasks such as sentiment classification.

3.4.3 KoBERT

KoBERT employs the SentencePiece tokenizer internally, which performs character-based subword segmentation rather than segmentation based on explicit morphological analysis. In this study, we adopted KoBERT’s built-in SentencePiece method without modification. This approach segments text according to the statistical properties of character sequences, without relying on linguistic rules. For example, the sentence “이 밥은 대박 맛있었다” may be segmented into space-based subwords [_이, _밥은, _대박, _맛있었다], and in some cases, further divided into smaller units [_이, _밥, 은, _대, 박, _맛, 있, 었, 다]. This method operates flexibly in languages like Korean, where particles, endings, and affixes are productively combined. It also has the advantage of effectively generalizing rare or unseen words that do not appear in the training data. The SentencePiece approach enables a certain degree of semantically meaningful segmentation without requiring morphological analysis, simplifies the preprocessing stage, and enhances the versatility of pre-trained language models. KoBERT’s masking strategy is similar to that of KoELECTRA; however, as noted earlier, it follows the traditional masked language modeling approach, in which tokens are masked and the model learns by predicting the original words.

3.4.4 KoNLPy

Using KoNLPy as a tokenizer for Korean sentiment analysis is particularly effective because it reflects the structural characteristics of the Korean language. Korean is an agglutinative language, meaning that words are formed by attaching various grammatical endings to a root. For example, the verb “맛있었습니다” consists of 맛 (taste), 있었 (past tense marker), and 습니다 (polite ending). While a basic tokenizer may treat this as a single word, KoNLPy decomposes it into smaller meaning-bearing units called morphemes. In addition, KoNLPy can extract the root adjective of a sentence and accurately identify grammatical elements such as negation, which can reverse the overall sentiment of a statement.

A key advantage of KoNLPy is its flexibility, as it provides access to multiple morphological analyzers. In this study, we used MeCab, a fast and efficient tokenizer for Korean. This choice was particularly suitable for our large-scale dataset, which includes over 3,000 diary entries and more than 100,000 words requiring efficient processing. MeCab’s high speed, low memory consumption, and accurate morphological analysis enable rapid preprocessing of large text corpora, making it well suited for Korean sentiment analysis tasks. Unlike methods such as SentencePiece, which segment text based on statistical patterns rather than linguistic structure, KoNLPy emphasizes grammatical and morphological information. It becomes even more powerful when integrated with deep learning architectures such as BERT, as it can refine and structurally organize text before it is processed by the model (Park and Cho, 2014).

3.5 Loss Function

3.5.1 Asymmetric Loss

Asymmetric Loss (ASL), proposed by Ben-Baruch et al. (2021), addresses the inherent class imbalance problem in multi-label classification by treating positive and negative labels differently

during training. In multi-label datasets, most instances contain only a small number of positive labels and a large number of negatives. As a result, standard loss functions such as Binary Cross-Entropy tend to overweight easy negative samples while underrepresenting rare positive ones. ASL introduces two key mechanisms. First, asymmetric focusing decouples the loss contributions of positive and negative samples by applying separate focusing parameters (γ^+ and γ^-). Second, probability shifting filters out extremely easy negative samples through hard thresholding, preventing them from contributing unnecessary gradients. Together, these mechanisms enable the model to focus on more informative and challenging samples, thereby improving performance while maintaining training efficiency and remaining architecture-agnostic. ASL has demonstrated strong empirical performance, achieving state-of-the-art results on benchmark multi-label datasets such as MS COCO, Pascal VOC, and Open Images without requiring additional model complexity or external resources (Ben-Baruch et al. 2021).

3.5.2 Binary Cross-Entropy Loss

In the referenced study, binary cross-entropy (BCE) loss is presented as the conventional baseline loss function for multi-label classification tasks. BCE loss treats each label prediction as an independent binary decision, calculating the loss for each label separately and then averaging the losses across all classes. The BCE loss is defined as:

$$L_{\text{BCE}} = -\frac{1}{n} \sum_{i=1}^n (Y_i \cdot \log \hat{Y}_i + (1 - Y_i) \cdot \log(1 - \hat{Y}_i)) \quad (1)$$

where $Y_i \in \{0, 1\}$ is the ground truth label, and $\hat{Y}_i \in [0, 1]$ is the predicted probability for that label. Although BCE loss is simple and widely used, the research highlights its major limitations, particularly in multi-label settings with severe class imbalance. Because most labels in such datasets are negative, BCE loss is often dominated by numerous easy negative samples. This imbalance can cause the model to underfit rare positive classes, as their contribution to the total loss is comparatively small. Consequently, models trained with BCE loss may achieve high overall accuracy while performing poorly on underrepresented labels. This makes BCE loss less suitable for tasks such as emotion classification, where multiple imbalanced labels frequently co-occur (Wu et al. 2020).

To better address class imbalance in multi-label classification tasks, researchers have proposed a weighted BCE (WBCE) loss function. In standard BCE loss, all labels are treated equally, which often leads to suboptimal performance on rare classes because the model tends to prioritize more frequent labels. To mitigate this issue, class-specific weights are incorporated into the BCE loss function. Higher weights are assigned to rare classes, while lower weights are given to more frequent ones. These weights are typically computed inversely proportional to class frequency. By amplifying the contribution of underrepresented classes to the overall loss, the model is encouraged to learn more robust and discriminative representations for those labels. The WBCE approach has been shown to significantly improve performance on imbalanced multi-label datasets. It is particularly effective for tasks such as emotion classification, where accurately detecting rare sentiments is crucial (Rezaei-Dastjerdehei et al. 2020).

3.6 FastText

FastText is a word embedding technique proposed by Facebook in 2016. It converts natural language into vector representations by leveraging internal word information. For example, the word “somewhere” can be represented as the sum of its subword n-gram vectors, such as “some” and “where”. A key advantage of FastText is its ability to capture morphological information through subword-level representations. However, this approach also has limitations. While meaningful subwords (e.g., “people” or “to”) may be extracted from a word, meaningless fragments such as “ram” can also be generated (Jeon, 2022). Furthermore, traditional word embedding methods often struggle with rare words and cannot handle out-of-vocabulary (OOV) terms that do not appear in the training corpus (Chae, 2019). To address these challenges, FastText

decomposes words into subword n-gram units and constructs each word vector as the sum of its corresponding subword vectors. This design improves the model’s ability to represent rare or morphologically complex words more effectively than standard word-level embedding approaches. FastText defines its scoring function using the softmax function to model the probability of a context word given a target word. For a target word w_t and a context word w_c , the conditional probability is:

$$P(w_c|w_t) = \frac{\exp(u_{w_c}^T v_{w_t})}{\sum_{w \in V} \exp(u_w^T v_{w_t})} \quad (2)$$

where v_{w_t} = vector representation of the target word, u_{w_c} = vector representation of the context word, and V = vocabulary. FastText represents a word as the sum of its character n-gram embeddings. So for FastText:

$$v_{w_t} = \sum_{g \in G_{w_t}} z_g \quad (3)$$

where G_{w_t} = set of character n-grams in word w_t and z_g = embedding of n-gram g .

In this study, we focus on the observation that FastText’s subword-based embedding method enables a certain degree of semantic unit segmentation without requiring explicit morphological analysis. This mechanism is conceptually similar to the subword segmentation strategies employed by BERT-based models, such as the SentencePiece tokenizer used in KoBERT and the WordPiece tokenizer adopted in KoELECTRA. This similarity suggests that FastText’s n-gram embedding structure may serve as an effective preprocessing approach even for Korean, a language characterized by complex verb inflections and frequent combinations of particles and endings. Accordingly, this study compares the performance of FastText’s n-gram embedding framework with the subword tokenization mechanisms of BERT-based models to assess their applicability to real-world Korean sentiment analysis tasks.

3.7 Auto-skip Embedding

In this study, we adopt an autoencoder-based label embedding method with skip concatenation (Figure 2) to reduce information loss in high-dimensional label spaces for multi-label classification (Kim and Kim, 2021). The skip-connected autoencoder introduces direct residual paths between corresponding encoder and decoder layers, addressing the limitations of traditional autoencoders (Hinton and Salakhutdinov, 2006). The authors demonstrated that deep architectures can effectively compress high-dimensional data into compact latent representations. However, when compressing one-hot label vectors, conventional autoencoders may lose important label correlations, increasing reconstruction error and degrading classification performance (He et al. 2015). To mitigate this issue, we incorporate skip (residual) connections—systematically introduced in ResNet—which help preserve information and stabilize training in deep networks. By directly linking inputs and intermediate representations across encoder and decoder layers, the proposed skip-connected autoencoder minimizes information loss during both compression and reconstruction of high-dimensional label vectors.

As shown in Figure 2, the proposed architecture first feeds the one-hot label vector through multiple encoder blocks to generate a low-dimensional latent representation. Each encoder block takes both the original one-hot input and the output of the preceding block, enabling the latent vector to more fully capture the original information. During decoding, the latent vector is reconstructed to its original dimensionality by summing the outputs of the corresponding encoder blocks through skip connections, thereby recovering details lost during compression. This bidirectional skip structure more effectively preserves nonlinear correlations among labels and significantly improves reconstruction accuracy.

Training is conducted in two stages, as shown in Figure 3. In the first stage, latent embeddings are learned using only the autoencoder by minimizing the mean squared error between the one-hot label vectors and the decoder outputs. In the second stage, the text classification model is trained to predict the latent vectors generated by the encoder in the first stage. The predicted latent vectors

are then passed through the decoder to produce the final multi-label outputs, thereby maximizing high-dimensional label reconstruction performance.

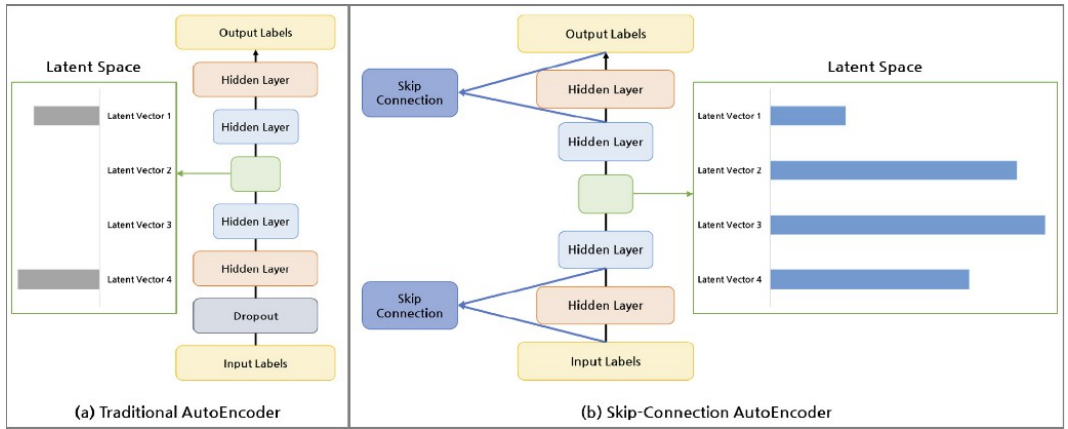


Figure 2. Autoencoder-based label embedding method with skip concatenation, adapted from Kim and Kim (2021).

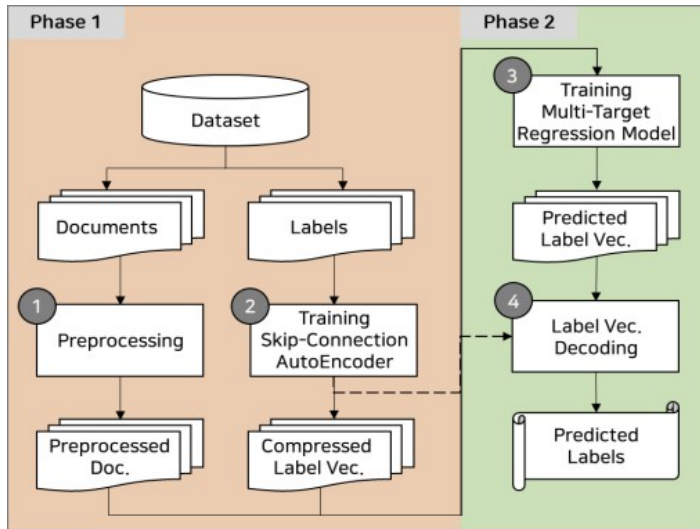


Figure 3. Training process of the skip-connection autoencoder-based label embedding method, adapted from Kim and Kim (2021).

4. Data source and Integration

4.1 Suitability of Diary Data for Emotion Classification

Diaries play a crucial role in emotion classification because they provide authentic, day-to-day reflections that capture subtle and complex emotional states in natural language. Unlike structured questionnaires or public social media posts, diary entries are typically written in private, allowing individuals to express their feelings more openly and comprehensively. The EmoDiary project demonstrates that combining diary writing with sentiment analysis through machine learning can enhance users' emotional self-awareness by identifying longitudinal emotional patterns (Soppari et al. 2025).

The goal of this study is to develop a machine learning model capable of analyzing emotions in Korean-language diary entries. By doing so, we aim to obtain deeper and more accurate insights into individuals' emotional well-being, leveraging the emotionally rich and naturally expressed

content found in private diary writing.

4.2 Rationale for Using Diverse Data Sources

In this study, we prioritized real-world diary data to train our emotion recognition model. However, diary entries are inherently private, rarely shared voluntarily, and raise ethical concerns related to privacy, anonymity, and informed consent. These factors imposed practical constraints on collecting a sufficiently large corpus.

Emotion labeling presented an additional challenge. Because emotion annotation involves subjective interpretation and often requires expertise in fields such as counseling psychology, constructing high-quality labeled data was difficult. Given our limited domain expertise, budget, and personnel, independently building a reliable annotated dataset was not feasible.

To address these limitations, we combined four complementary datasets. First, we utilized the KOTE dataset (Jeon, Lee, & Kim, 2022), which contains approximately 50,000 online comments labeled across 43 emotion categories. However, its short, comment-style format restricts the expression of complex emotions. Second, we incorporated artificially generated emotion sentences to enhance lexical diversity and alleviate class imbalance. Third, we included blog posts from NAVER to obtain more natural narrative contexts and stylistic similarities to diary writing, particularly expressions related to mental health. Finally, we collected a small set of real diary entries internally to strengthen domain consistency.

This data construction strategy mitigated category sparsity, improved stylistic diversity and authenticity, and supported the development of a model capable of capturing nuanced emotional expressions in diary texts.

To further address data sparsity and labeling constraints, we adopted a multi-level semi-supervised learning framework. Specifically, we employed pseudo-labeling to iteratively annotate unlabeled data and expand the training set, enabling effective model training with limited labeled resources (Yu et al. 2021; Yang et al. 2023). We also explored complementary approaches, including multi-level semi-supervised learning, human-annotated synthetic diaries, and synthetic data generation using large language models such as ChatGPT, in addition to leveraging the KOTE dataset and web-scraped data described in the following sections.

4.3 Description of Data Sources

4.3.1 KOTE Dataset

To support our emotion recognition goals, we utilize the KOTE dataset, a benchmark multi-label corpus widely used in Korean-language NLP research. The KOTE dataset serves as the foundation for our multi-label emotion analysis experiments. It comprises 50,000 Korean-language comments collected from 12 diverse online platforms—including news sites, blogs, and social media—providing broad linguistic and contextual coverage. KOTE employs a fine-grained 43-category emotion taxonomy to capture nuanced emotional states. To ensure annotation quality, each comment was labeled independently by five native Korean speakers, and the final emotion scores were aggregated from these individual ratings. Following the original methodology, multi-rater scores were min-max normalized and binarized using a threshold of 0.2, yielding an average of 7.91 active emotion labels per comment. The dataset is split into standard research partitions: 80% for training, 10% for validation, and 10% for testing, facilitating model development and unbiased performance evaluation (Jeon, Lee, and Kim, 2022).

In this study, we examine KOTE's label distribution, co-occurrence patterns, and sentence-length characteristics to identify potential dataset biases and structural patterns that could influence model performance. These analyses provide a critical foundation for subsequent modeling decisions and evaluation strategies (Jeon, Lee, and Kim, 2022).

As shown in Figure 4, bar plot analysis reveals a pronounced imbalance in emotion frequencies. Dominant categories such as Disappointment, Complaint, and Absurdity occur frequently, while emotions like Guilt, Fear, and Defeat/Self-hate appear only rarely.

As shown in Figure 5, heat-map visualization shows frequent co-occurrences among related emotions, particularly between negative states such as Irritation and Anger, and between Anxiety/Worry and Sadness. While these patterns reflect psychological realism in the annotations,

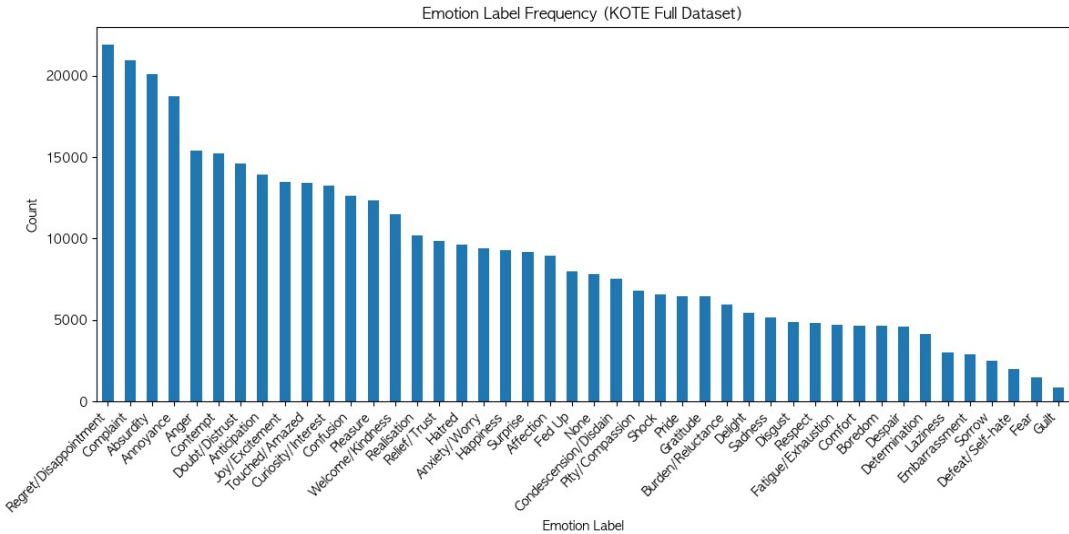


Figure 4. Histogram of the emotion labels in the KOTE dataset.

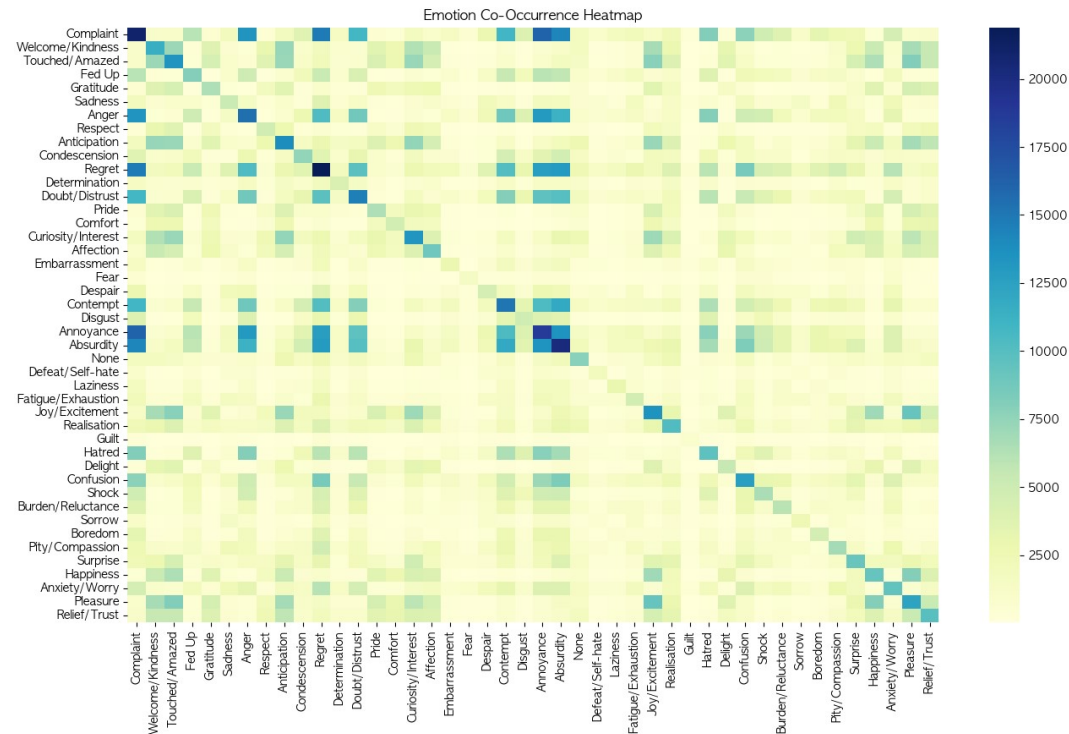


Figure 5. Co-occurrences among related emotion labels in the KOTE dataset.

As shown in Figure 6, sentence length analysis shows that most entries in the dataset are short to medium in length, typically between 30 and 50 characters. This distribution indicates that NLP models optimized for short-text processing may achieve greater efficiency.

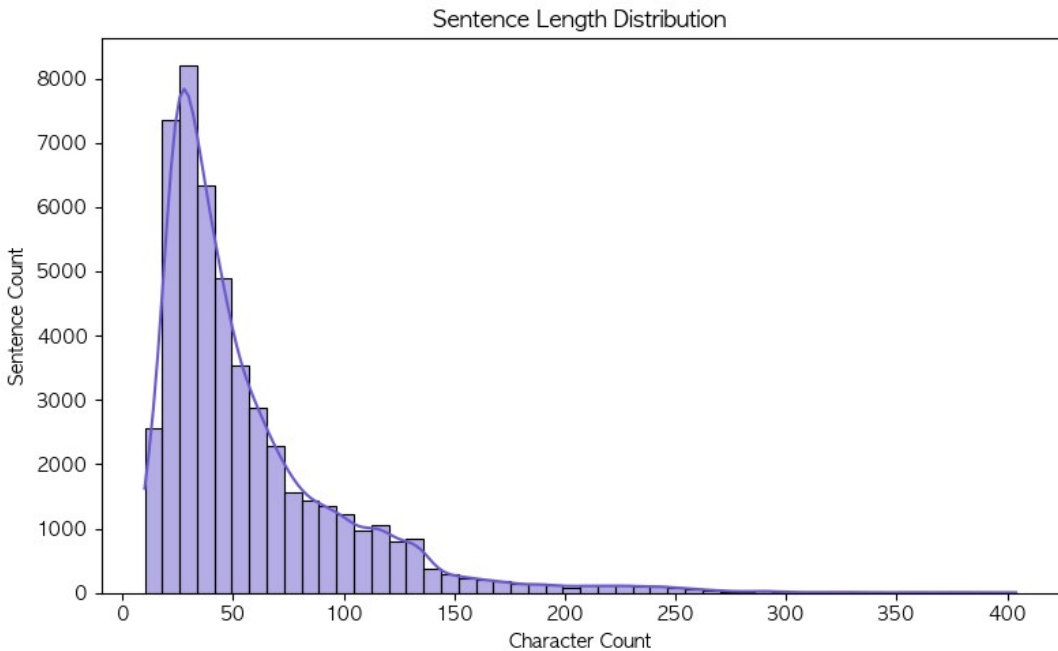


Figure 6. Histogram of sentence lengths in the KOTE dataset.

4.3.2 Synthetic Data Generation with ChatGPT

Synthetic data—algorithmically generated text that mimics real language—provides a privacy-preserving method to expand sparse or sensitive NLP corpora. By creating fictional diary entries, sentiment-laden sentences, or dialogues, researchers can quickly produce large, automatically labeled datasets, bypassing data-access restrictions and costly manual annotation. Studies show that models trained on synthetic corpora can match or outperform those trained on original data: for example, a Synthetic Data Vault benchmark found equal or better predictive accuracy in 11 of 15 tasks, while ChatGPT-generated augmentation improved low-resource performance to 78–91% across SST-2, SNIPS, and TREC datasets, surpassing back-translation, EDA, and GPT-2 baselines—even achieving competitive results when trained solely on synthetic samples (Patki et al. 2016; Ubani, Polat, and Nielsen, 2023). These findings demonstrate the practicality of synthetic data for robust model training under data-scarce or privacy-constrained conditions.

To overcome limitations of time, budget, and data availability, we generated synthetic diary entries using ChatGPT. These entries closely mimic real diary writing in structure and tone, making them suitable substitutes for authentic data. Following recent evidence supporting synthetic data for emotion classification, we created 400 entries—one for each day in a simulated daily log. To ensure diversity, each team member produced 100 entries in collaboration with ChatGPT, designing unique fictional characters with distinct lifestyles, personalities, and emotional tendencies. This strategy simulated a wide range of emotional contexts and writing styles, while using ChatGPT as a neutral tool helped reduce bias, maintain consistency, and preserve the emotional variability required for training a robust emotion recognition model.

4.3.3 Written Diaries

In addition to publicly available and synthetic sources, we included a small set of hand-written diaries contributed by team members. These entries provide authentic, long-form, introspective narratives that closely reflect the linguistic style and emotional depth of personal diary writing, offering valuable domain-specific data. However, to address gaps in emotion coverage and class imbalance—especially for low-frequency emotions—additional data was needed. Synthetic data generated specifically for model training presents a practical and ethical solution. Recent studies show that combining large language model (LLM)-generated content with real data can

significantly improve recall for minority emotion classes while maintaining overall model stability (Jeon, Lee, and Kim, 2022).

To overcome limitations in data availability and labeling, we created a synthetic dataset of Korean diary entries. We developed fictional diary personas representing different MBTI personality types, each contributing 100 entries to capture diverse emotional expressions and internal thought patterns. For example, one persona, an ENFP high school student named “Rapunzel,” writes expressive, imaginative entries rich in spontaneous reflection and introspective creativity. By designing varied personality-based diaries, we simulated real emotional complexity while ensuring privacy and ethical compliance. This approach allows for balanced representation across personalities, enables pre-labeling of emotional content, and provides training data that closely mirrors authentic personal reflection for robust emotion recognition model development.

4.3.4 Naver Blogs

Park et al. (2022) used Naver blog posts as legally public sources in their studies, demonstrating both methodological feasibility and ethical permissibility when proper anonymization is applied. To complement our synthetic diary data, we explored Naver Blog as a source of real-world emotional writing. As South Korea’s largest search portal, Naver’s blog platform hosts diverse user-generated content, including travel journals, personal reflections, and daily diaries, often reflecting genuine emotional states—making it valuable for training emotion analysis models. However, directly scraping blog content carries ethical and legal risks, particularly copyright infringement. Many posts include explicit restrictions such as “Author attribution required – No commercial use – No modification”, preventing their use or modification without consent.

To address this, we developed a custom Python web scraping script that processes each Naver blog link by extracting the blog ID and post ID. The script navigates to the blog’s landing page, locates the full post through the embedded iframe, and checks the HTML for copyright protection notices. Only posts without such labels were extracted, ensuring that our dataset includes content that is both legally and ethically usable. By automating this filtering process, we respect content creators’ rights while expanding our dataset with authentic, diverse emotional expressions written in natural Korean.

4.4 Data Integration Strategy

To effectively train our emotion recognition model, we employ a multi-stage, data-rich strategy that integrates small-scale labeled datasets with large-scale unlabeled or semi-labeled sources. We begin with a core labeled dataset—such as the KOTE dataset—which provides a reliable baseline for initial training and validation. Using this high-quality labeled data, the model can learn foundational patterns of emotion. Once trained on KOTE, the model can be used to weakly label additional data sources, including synthetic MBTI diary entries generated with ChatGPT, ethically scraped blog content from Naver, and other synthetic diary-style texts. This semi-supervised learning approach allows us to expand our training dataset significantly without the need for extensive manual annotation. Leveraging the pre-trained model, these new texts are annotated with emotional categories and used to further fine-tune the model. This pipeline not only increases both the quantity and diversity of training data but also improves the model’s ability to generalize across different writing styles, emotional tones, and contexts. Consequently, it enhances performance, particularly in low-resource or sensitive language domains such as Korean diary entries.

4.5 Enhancing AI Model Accuracy with Semi-Supervised Learning through Pseudo-Labeling and Data Augmentation

Training accurate AI models often requires large amounts of labeled data, which can be costly and labor-intensive. This paper introduces a novel approach that integrates semi-supervised learning with data augmentation and pseudo-labeling to improve model accuracy, particularly in settings with limited labeled data. Label scarcity remains a major challenge in AI model development, but recent studies have highlighted data augmentation and pseudo-labeling as effective solutions. In

this study, we explore a combined strategy that generates additional labeled data from unlabeled datasets, enhancing model performance through iterative training processes (Yu et al. 2021; Yang et al. 2023).

5. Experimental Design and Evaluation

5.1 Experimental Objective

This study aims to improve the performance of a multi-label emotion classification model for Korean emotions. Because human emotions are diverse and complex, representing an individual’s emotional state with a single label is often insufficient. Previous research on Korean multi-label emotion classification has been limited by data imbalance, reliance on general loss functions, and the use of single-model architectures.

This study addresses the following research questions:

1. To what extent can a hybrid structure of Korean BERT models improve multi-label classification accuracy compared to conventional single-model architectures?
2. How much do optimized loss functions and Korean-specific tokenizers contribute to performance improvements?
3. Can the data imbalance problem be effectively addressed through class consolidation, deletion, and the addition of newly collected data?
4. What is the optimal model combination when using ensemble techniques?

To answer these questions, we first compared the performance of KoBERT, KoELECTRA, and KcBERT to identify the most effective pre-trained model. We then proposed an optimized loss function and a Korean-specific tokenizer to maximize performance. To tackle data imbalance, we reduced the original 43 sentiment classes to 37 by consolidating or removing redundant or underrepresented classes, and expanded the dataset with additional labeled data. Moreover, we explored a hybrid architecture incorporating FastText embeddings and an auto-skip mechanism, and evaluated performance improvements using threshold optimization and ensemble strategies. Overall, our approach demonstrates substantial improvements in multi-label emotion classification for Korean, offering valuable insights into model architecture, data management, and optimization strategies in Korean NLP.

5.2 Experimental Environment and Hyper Parameter Setting

All experiments in this study were conducted on Linux-based workstations, with the primary hardware and software environment as described below. Model implementation and experiments were performed using PyTorch, the HuggingFace Transformers library, and various Python packages including scikit-learn, NumPy, and pandas. GPU acceleration was provided by the NVIDIA CUDA environment, and a single NVIDIA RTX 3090 GPU was used for the main experiments.

The key hyperparameters were set as follows. The training batch size was 16 to balance GPU memory constraints and convergence speed, while the evaluation batch size was increased to 32 for efficiency. Training was limited to 100 epochs, with early stopping applied to prevent overfitting. Specifically, the minimum improvement in validation loss (`min_delta`) was set to 0.005, and training stopped automatically if no improvement occurred for 10 consecutive evaluations (`patience` = 10). A linear learning rate scheduler with warm-up was applied, gradually increasing the learning rate over 10% of the total training steps (`warmup_ratio` = 0.1) and then decreasing it for stable convergence. Optimization was performed using the AdamW optimizer with a weight decay of 0.01. Mixed-precision training (`fp16`) was enabled to improve computational efficiency and reduce memory usage. The number of data loader workers was set to 4 to improve data input/output efficiency. For experimental comparisons, pre-trained language models, tokenizers, label embedding methods, and hybrid network structures were adjusted according to different configurations. Specifically, we explored variations including the

KoELECTRA WordPiece tokenizer, the Mecab morphological analyzer, FastText-based label representations, and hybrid network structures combining [CLS] vector 300-dimensional projections with label embeddings. Detailed configurations for each experimental setup are described alongside the results.

5.3 Nested Cross-Validation and Multi-Metric Evaluation

To reliably evaluate the generalization performance of multi-label classification models, we employed a combination of Multilabel Stratified 5-Fold cross-validation and internal hold-out validation. First, the entire dataset was partitioned into five folds using the Multilabel Stratified K-Fold technique, and five experimental iterations were performed, with each fold serving as a test set once. Within each training fold, 10% of the data was randomly selected as an internal validation set, and the remaining 90% was used for actual training. The internal validation set monitored model performance across multiple metrics per epoch and enabled the application of early stopping to prevent overfitting.

This dual validation structure offers several advantages. First, it mitigates imbalance in multilabel distributions, ensuring each sentiment label is proportionally represented across folds. Second, the external 5-fold experiments reduce performance variance on the test set, providing a more reliable estimate of the model’s generalization ability. Third, internal hold-out validation effectively controls overfitting by automatically identifying the optimal training duration (Hinton and Salakhutdinov, 2006). This approach is particularly valuable for multi-label sentiment analysis, where labels are interdependent and data distributions are complex (Arlot and Celisse, 2010).

During training, the folds were divided into training and validation sets to optimize model parameters and determine early stopping criteria. The best-performing model from each fold was then applied to the test set for final evaluation. This double cross-validation structure minimizes accidental bias from a single split and ensures consistent model behavior across different data distributions.

Given the complexity of multi-label sentiment classification, model performance was assessed using multiple metrics:

- **Macro-F1 Score:** The unweighted mean of F1 scores across all classes, ensuring sparse sentiment labels are equally represented.
- **Hamming Loss:** Measures the percentage of incorrectly predicted labels, providing an intuitive sense of multi-label error.
- **ROC-AUC:** Evaluates binary classification performance per class and averages across all classes.

Additional metrics, including accuracy, sample-level precision/recall/F1 (F1_samples), macro/micro metrics, and the Matthews correlation coefficient (MCC), were also calculated to enhance result reliability and interpretability. Macro-F1 was chosen as the primary evaluation metric to fairly assess performance across all emotion labels in an unbalanced dataset. By using this multi-metric, dual-validation approach, we comprehensively examined the stability and balance of the model’s overall classification ability, as well as the performance differences among individual emotion labels.

5.4 Model Selection

In this section, we systematically compare the multi-label emotion classification performance of several pre-trained Korean language models. Model effectiveness was evaluated using key classification metrics, including Macro-F1 Score, AUC, Hamming Loss, and sample-level F1 Score, Precision, and Recall. Table 1 summarizes the core results for the three models, including KoELECTRA, KcBERT, and KoBERT under identical experimental conditions.

Table 1. Comparison of multi-label classification performance among the pre-trained KoELECTRA, KcBERT, and KoBERT models

Metric	KoELECTRA	KcBERT	KoBERT
Loss	1.0742	1.1241	1.1218
Exact Match Accuracy	0.0070	0.0060	0.0062
Accuracy	0.8550	0.8490	0.8513
Precision	0.6126	0.5946	0.6083
Recall	0.8404	0.8287	0.8285
F1-score	0.6932	0.6781	0.6845
Hamming Loss	0.1450	0.1510	0.1487
Macro Precision	0.5264	0.5112	0.5224
Macro Recall	0.4980	0.4675	0.4725
Macro F1-score	0.5077	0.4832	0.4929
AUC	0.8253	0.8040	0.8174
MCC	0.4161	0.3890	0.4010

The experimental comparison of KoELECTRA, KcBERT, and KoBERT provides strong empirical evidence for selecting KoELECTRA as the optimal backbone for multi-label emotion classification in Korean. Across all primary evaluation metrics, KoELECTRA demonstrated consistently superior performance. Specifically, it achieved a macro-F1 score of 0.5077, substantially higher than KcBERT (0.4832) and KoBERT (0.4929). This advantage was mirrored in other metrics: KoELECTRA attained a sample-wise F1 score of 0.6932, outperforming KcBERT (0.6781) and KoBERT (0.6845), along with the highest precision (0.6126), recall (0.8404), and accuracy (0.8550).

In terms of class discrimination, KoELECTRA led with an AUC of 0.8253, compared to 0.8040 for KcBERT and 0.8174 for KoBERT. Its Hamming Loss was the lowest at 0.1450, indicating fewer multi-label misclassifications than KcBERT (0.1510) and KoBERT (0.1487). KoELECTRA also demonstrated greater stability at the class level, achieving macro-precision and macro-recall of 0.5264 and 0.4980, respectively—higher than KoBERT (0.5224 and 0.4725) and KcBERT (0.5112 and 0.4675).

Although KoBERT showed slightly better sample-wise precision and F1 than KcBERT, these gains were insufficient to offset its lower performance on key aggregate metrics. In contrast, KoELECTRA consistently outperformed both alternatives across macro-level, sample-level, and class-level evaluations. These results were observed under identical experimental conditions—including data preprocessing, training strategies, and evaluation protocols—reinforcing the robustness of the comparison.

Given its consistently higher macro-F1 score, superior AUC, lower Hamming Loss, and overall stability across metrics, KoELECTRA emerges as the most effective model. These findings provide a solid quantitative foundation for selecting KoELECTRA as the primary architecture for subsequent hybrid and advanced modeling approaches in this study.

5.5 Tokenizer Optimization

An in-depth analysis of sample-based precision, sample-based F1 score, and Hamming Loss reveals consistent effects linked to the choice of tokenizer, independent of model architecture or loss function. While the magnitude of these changes is generally small, the impact of the Mecab tokenizer is evident across different experimental configurations.

For the KcBERT model with standard BCE loss, using Mecab slightly increased sample-based precision (from 0.5946 to 0.5972) and F1 score (from 0.6781 to 0.6820), while marginally reducing Hamming Loss (from 0.1510 to 0.1506). In the case of KoELECTRA combined with ASL, Mecab led to a more notable improvement in sample-based precision (from 0.5896 to 0.6145), a slight decrease in F1 score (from 0.6974 to 0.6942), and a significant reduction in Hamming Loss (from 0.1522 to 0.1442).

Although the absolute changes vary across model–loss combinations, these examples illustrate a consistent trend: adopting Mecab reduces multi-label misclassification errors and, in several cases, enhances the model’s ability to identify relevant emotion labels within individual samples. The modest improvements in sample-based precision and F1 for KcBERT+BCE suggest that Mecab helps extract more linguistically meaningful features from Korean text. This morphological analysis is particularly valuable in Korean, where complex word boundaries and grammatical structures require context-sensitive disambiguation to capture emotional nuances (Shim and Yang, 2002).

The observed reductions in Hamming Loss further indicate that Mecab improves the reliability of multi-label predictions by decreasing false or redundant label assignments. Even when sample-based F1 does not consistently increase, the persistent reduction in Hamming Loss across models demonstrates that morphological tokenization contributes to a minor but robust improvement in multi-label emotion classification performance.

5.6 Comparison of Loss Functions after Class Reduction

5.6.1 Class Reduction

To address the challenge of class imbalance and improve model robustness, we conducted a systematic class reduction and consolidation process. For each combination of the three base models and three tokenizers, we computed recall, precision, and F1-score for every emotion class across the five cross-validation folds. By aggregating these results, we calculated the mean F1-score for each class and ranked all classes accordingly. Figure 7 shows the plot of average F1-score over each class. Classes with an average F1-score below 0.35 were identified as underperforming categories.

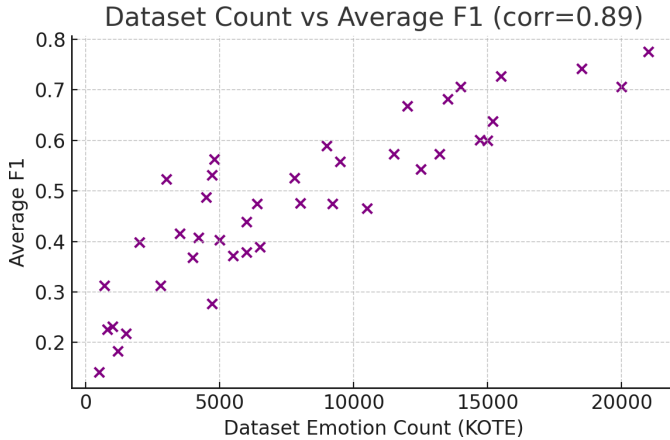


Figure 7. Average F1-score over each emotion label with the number of samples in the KOTE dataset.

Reducing underperforming or redundant classes was expected to provide several benefits. By merging semantically similar classes and removing those with consistently low predictability, we aimed to reduce data sparsity and label ambiguity, facilitate more stable learning, mitigate overfitting to rare labels, and enhance the reliability and interpretability of multi-label predictions, particularly in class-imbalanced datasets.

Eight emotions were selected for review based on low performance: Despair (0.3128), Fear (0.3127), Burden/Reluctance (0.2765), Sorrow (0.2321), Defeat/Self-hate (0.2259), Laziness (0.2175), Embarrassment (0.1827), and Guilt (0.1418). Among these, some were merged with semantically similar or higher-performing classes, while others were removed when consolidation was not feasible. The merging scheme was as follows:

- Despair and Sorrow → Sadness
- Defeat/Self-hate and Guilt → Self-blame

- Burden/Reluctance and Laziness → Fatigue/Exhaustion
- Fear unified with Anxiety/Worry → Anxiety-Fear
- Embarrassment, which showed particularly low performance, was eliminated

This consolidation reduced the total number of target emotions from 43 to 37, creating a more balanced and robust classification setting, as shown in Figure 8. After remapping and merging, we observed a decrease in the number of emotions per diary entry; however, the top three most frequent emotions in the original KOTE dataset—Regret, Complain, and Fatigue—remained the most common in the remapped dataset. This suggests that diary writers tended to express negative emotions more frequently, even after class consolidation. Furthermore, in the original dataset, Fear and Guilt were among the least frequently occurring emotions. After merging them during class consolidation, Delight became the least frequent emotion in the remapped dataset. Additionally, we observed a substantial frequency drop between Sadness and Fed-up, highlighting a pronounced imbalance even after consolidation.

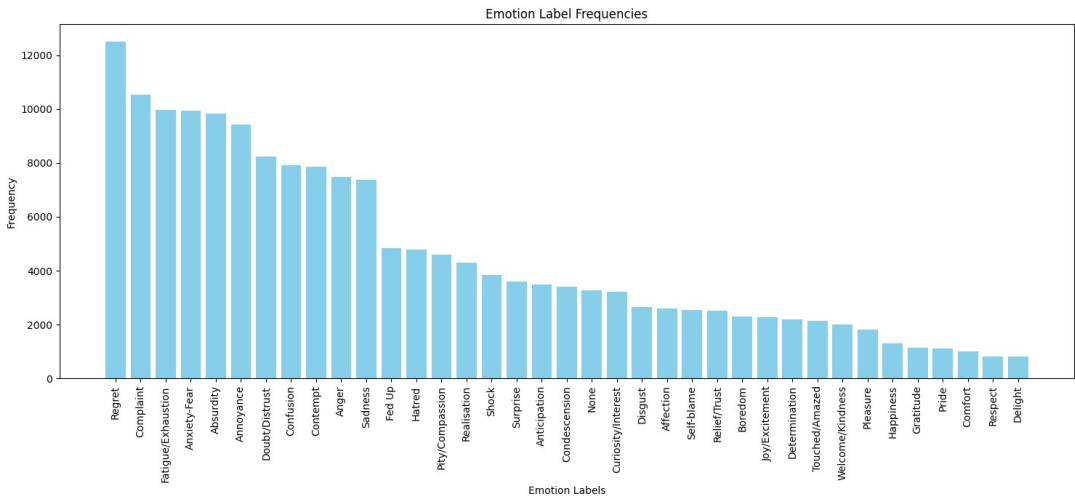


Figure 8. Histogram of emotion labels after class reduction.

5.6.2 Loss Function Comparison

As previously noted, standard BCE loss is not ideally suited for multi-label classification tasks, particularly under conditions of severe class imbalance. To address this limitation, we explored two alternative approaches: ASL and WBCE, both of which have demonstrated effectiveness in multi-label learning. These loss functions were systematically applied to the best-performing architecture identified in earlier experiments—namely, KoELECTRA with the Mecab tokenizer and the reduced set of 37 emotion classes. This setup enabled a rigorous evaluation of the impact of loss function optimization on multi-label emotion classification performance.

Reducing the total number of emotion classes to 37 resulted in noticeable improvements across all primary evaluation metrics, highlighting the positive effect of mitigating class imbalance and label sparsity. Within this more balanced experimental framework, we conducted a direct comparison between ASL and WBCE.

The empirical results show that WBCE consistently outperformed ASL across all major performance indicators. WBCE achieved an accuracy of 0.8611, compared to 0.8583 for ASL. Sample-wise precision, recall, and F1 score were also higher under WBCE (0.6397, 0.8665, and 0.7292, respectively) than under ASL (0.6268, 0.8547, and 0.7252). These improvements indicate that WBCE enhances the model’s ability to correctly identify relevant emotion labels while reducing both false positives and false negatives.

Macro-averaged metrics—particularly important in imbalanced multi-label settings—also favored WBCE. Specifically, WBCE achieved a macro-precision of 0.5242, macro-recall of 0.6234, and macro-F1 of 0.5678, exceeding the corresponding ASL values of 0.5216, 0.6080, and

0.5593. These gains are meaningful, as macro-level metrics reflect performance across all classes, including underrepresented ones.

Hamming Loss decreased from 0.1417 (ASL) to 0.1389 (WBCE), indicating fewer misclassified labels overall. Additionally, WBCE demonstrated stronger discriminative ability, achieving a higher AUC (0.8392 vs. 0.8241) and a higher MCC (0.4532 vs. 0.4437). The improvement in MCC further suggests enhanced overall classification reliability despite residual imbalance.

Taken together, these findings demonstrate that incorporating class weights into the BCE loss function not only improves aggregate performance metrics but also promotes more stable and reliable multi-label emotion classification in settings characterized by class imbalance and label reduction. The results support WBCE as the preferred loss function for complex, real-world multi-label classification tasks in natural language processing.

5.7 Data Augmentation and Expansion

5.7.1 Semi-Supervised Learning Strategy via Pseudo-Labeling

To address the limitations of labeled data scarcity and enhance the model’s generalization capability, we adopted a semi-supervised learning framework based on pseudo-labeling. Although expert-annotated data is ideal, it is often costly and time-consuming to obtain. To mitigate this constraint, we utilized our best-performing configuration—KoELECTRA + Mecab + WBCE—as a Teacher Model to generate pseudo-labels for newly collected unlabeled text entries.

The theoretical foundation of this approach lies in the principle of entropy minimization. Foundational work in semi-supervised learning suggests that training a model on its own high-confidence predictions reduces the conditional entropy of class probabilities on unlabeled data (Grandvalet and Bengio, 2004). This mechanism effectively pushes decision boundaries toward low-density regions, thereby increasing class separation and reducing ambiguity (Lee, 2013).

In the context of multi-label emotion classification, maintaining the complex correlations and co-occurrence patterns among emotion labels is particularly important. Recent studies indicate that robust encoders and large language models can preserve such semantic dependencies when generating pseudo-labels, even without human annotation (Mousavi et al. 2023; Han et al. 2025). Since our Teacher Model had already learned the intricate relationships among the 37 emotion categories from the KOTE dataset, the pseudo-labels generated for new data reflect these learned co-occurrence structures rather than arbitrary assignments. This approach expands the training distribution while preserving domain consistency (Rizve et al. 2021).

The practical effectiveness of this strategy is supported by comparative performance analysis before and after data expansion. Prior to expansion, the optimized model achieved a macro recall of 0.6234 and an AUC of 0.8392. After incorporating pseudo-labeled synthetic diary entries and real-world blog data, macro recall increased substantially to 0.6658, representing an improvement of approximately 6.8%, while AUC slightly increased to 0.8404.

The marked improvement in recall indicates enhanced sensitivity to diverse and subtle emotional expressions that were underrepresented in the original KOTE dataset. Importantly, the stability—and slight improvement—of AUC demonstrates that incorporating unstructured “wild” data did not degrade the model’s discriminative capability. This suggests that the generated pseudo-labels were of sufficient quality to broaden the model’s generalization boundaries without introducing excessive noise.

Consequently, the trained Teacher Model was used to annotate additional unlabeled texts, including synthetic diary entries and real-world blog posts. These newly labeled samples were merged with the original KOTE dataset, resulting in a substantially expanded and more diverse training corpus, as shown in Figure 9.

As illustrated in Figure 10, the final integrated dataset comprises three primary sources, totaling approximately 70,000 entries:

- KOTE Dataset (71.7%, 50,000 entries)
This dataset serves as the foundational data source, providing a reliable, expert-annotated baseline for Korean multi-label emotion classification.

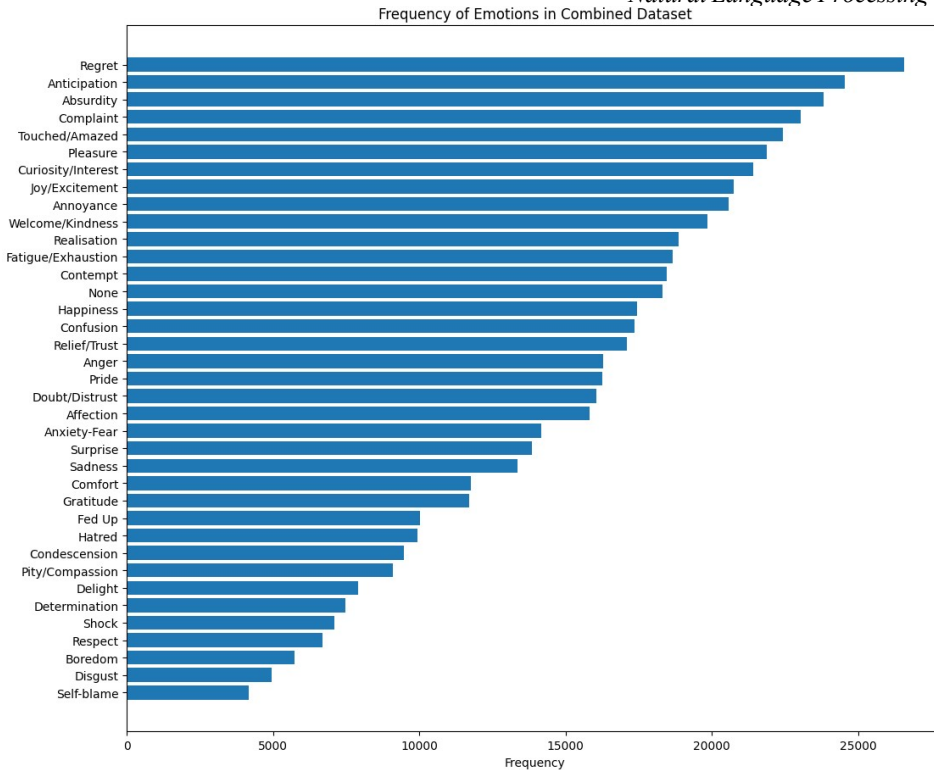


Figure 9. Histogram of emotion labels in the combined dataset.

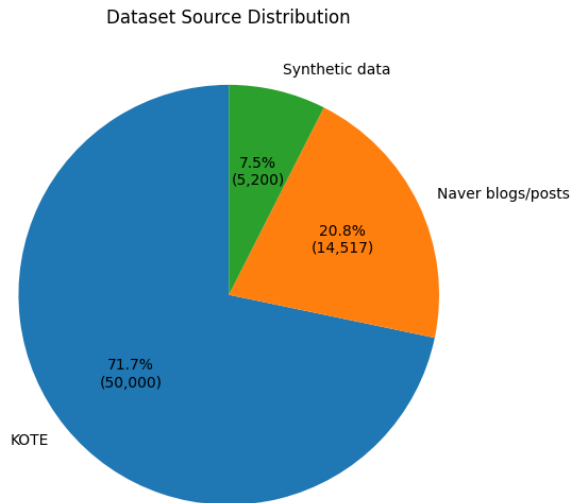


Figure 10. Structure of the combined dataset.

- **Synthetic Diary Data (7.5%, 5,200 entries)**
Constructed by our research team, this subset includes both ChatGPT-generated text and manually written diary samples. This addition was essential for increasing linguistic diversity and capturing the introspective, narrative style characteristic of personal diaries, which differs substantially from the shorter, comment-style texts in KOTE.
- **Naver Blogs and Posts (20.8%, 14,517 entries)**
Collected from South Korea’s largest web portal, this portion reflects naturalistic and informal Korean expressions, representing the “wild” data distributions commonly observed in real-

All heterogeneous data sources were segmented and annotated using the unified 37-emotion scheme derived from the KOTE dataset. By employing the KoELECTRA + Mecab + WBCE model for pseudo-labeling, we maintained consistent labeling quality across these diverse domains.

This integration strategy not only alleviated data sparsity but also expanded the overall data distribution, enabling the final model to learn from a broader range of linguistic patterns and emotional contexts than would have been possible using the KOTE dataset alone. As a result, the model achieved improved robustness and generalization in real-world multi-label emotion classification scenarios.

5.7.2 Model Performance Comparison with the Expanded Dataset

To ensure direct comparability with the original KOTE benchmark, the classification threshold was fixed at 0.3, in accordance with the protocol described in the KOTE paper. We evaluated three configurations built upon the KoELECTRA + Mecab + WBCE framework: (1) the original baseline model, (2) a variant augmented with FastText embeddings, and (3) a variant incorporating skip connections. All models were trained and evaluated under identical experimental conditions using the unified set of 37 emotion classes, enabling a controlled and fair comparison of the architectural modifications. Table 2 shows their comparison results.

Table 2. Performance comparison of skip-connection-based, baseline, and FastText-based models

Metric	Skip-Connection	Baseline	FastText
Accuracy	0.8005	0.8110	0.8097
Precision (samples)	0.5402	0.5551	0.5530
Recall (samples)	0.7618	0.7430	0.7447
F1 (samples)	0.6021	0.6055	0.6050
Macro Precision	0.4798	0.4995	0.4985
Macro Recall	0.6901	0.6658	0.6650
Macro F1	0.5638	0.5686	0.5672
Hamming Loss	0.1995	0.1889	0.1903
ROC-AUC	0.8409	0.8404	0.8394
MCC	0.4450	0.4516	0.4494

A detailed examination of the evaluation metrics indicates that, although measurable differences exist among the baseline, FastText embedding, and skip-connection configurations, these differences are generally moderate rather than substantial. The baseline configuration—KoELECTRA + Mecab + WBCE, using a threshold of 0.3—achieved a sample-based F1 score of 0.6055, macro F1 of 0.5686, macro precision of 0.4995, and an MCC of 0.4516. These values were either the highest or nearly the highest among the evaluated variants. Moreover, the baseline recorded the lowest Hamming Loss (0.1889), indicating the fewest overall label misclassifications.

The FastText embedding variant produced results that were nearly indistinguishable from the baseline: a sample-based F1 of 0.6050, macro F1 of 0.5672, macro precision of 0.4985, and MCC of 0.4494. The differences in F1 and precision metrics were within 0.002, suggesting minimal practical divergence. Hamming Loss was slightly higher (0.1903), and accuracy (0.8097) was marginally below the baseline (0.8110). Overall, these results suggest that incorporating FastText embeddings neither substantially improves nor meaningfully degrades performance relative to the baseline.

The skip-connection configuration exhibited a distinct trade-off pattern. It achieved the highest macro recall (0.6901) and sample recall (0.7618), surpassing both the baseline (0.6658, 0.7430) and the FastText variant (0.6650, 0.7447). This indicates improved sensitivity in detecting relevant

emotion labels and a reduced likelihood of false negatives. However, this gain in recall was accompanied by declines in precision and overall balance. Macro precision decreased to 0.4798, and sample-based precision fell to 0.5402, both lower than the other configurations. Correspondingly, macro F1 (0.5638) and sample F1 (0.6021) were slightly reduced, while Hamming Loss increased to 0.1995, the highest among the three. Accuracy (0.8005) and MCC (0.4450) also showed a modest decline relative to the baseline and FastText models.

Overall, the three configurations demonstrate comparable performance, with aggregate metric differences remaining relatively small. The baseline configuration offers the most balanced performance, particularly in terms of precision, F1 score, MCC, and Hamming Loss. The FastText variant provides nearly equivalent results, suggesting that additional semantic embedding contributes limited incremental benefit under the current setup. The skip-connection model, while improving recall, introduces a trade-off in precision and overall stability.

Therefore, no configuration emerges as universally superior. Model selection should be guided by application-specific priorities. If balanced performance and minimal overall error are desired, the baseline configuration is an appropriate default. The FastText variant serves as a comparable alternative, while the skip-connection model may be preferable in scenarios where maximizing recall and minimizing false negatives are more critical than maintaining high precision.

The key insight is that, under the present dataset and experimental conditions, hybrid architectural modifications yield only marginal performance differences. Consequently, final model selection should be driven by downstream task requirements rather than by a decisive superiority in aggregate metrics.

5.8 Threshold Optimization

After training the three final configurations—baseline (KoELECTRA + Mecab + WBCE), the FastText embedding variant, and the skip-connection variant—we conducted decision threshold optimization for each model. Although earlier experiments employed a fixed threshold of 0.3 to ensure comparability with the KOTE benchmark, additional tuning was performed to further balance precision and recall in the multi-label setting.

For each configuration, we systematically evaluated a range of threshold values and monitored key performance metrics, including macro F1, sample-based F1, precision, recall, and Hamming Loss. This procedure allowed us to examine how varying the decision boundary influenced the trade-off between false positives and false negatives. The optimal threshold for each model was selected through analysis of the precision–recall (PR) curve, which provides a direct visualization of the precision–recall trade-off across different threshold values. By identifying the point that maximized macro F1 while maintaining acceptable precision–recall balance, we determined the most appropriate operating threshold for each configuration. Figure 11 shows some representative PR curve examples from the baseline model to illustrate the threshold selection process.

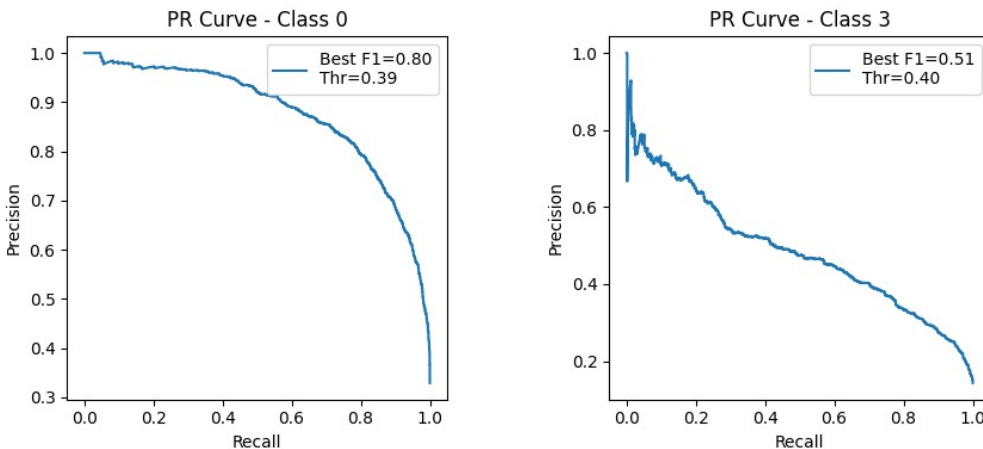


Figure 11. PR Curves of the baseline model for class 0 and class 3 for threshold selection.

As shown in Tabel 3, threshold optimization produced consistent improvements across all three configurations—the baseline, FastText embedding, and skip-connection models. After tuning the decision threshold, the macro F1 score increased from 0.6270 to 0.6498 in the baseline model, from 0.6271 to 0.6503 in the FastText variant, and from 0.6121 to 0.6439 in the skip-connection configuration. Although these gains appear moderate, they reflect a more effective balance between precision and recall across emotion classes, underscoring the importance of threshold tuning in multi-label, class-imbalanced settings.

Table 3. Performance comparison of skip-connection-based, baseline, and FastText-based models before and after threshold optimization

Threshold Optimization Before/After	Macro F1	Macro Precision	Macro Recall	AUC	Hamming Loss
Baseline (before)	0.627027	0.521556	0.798748	0.889282	0.1888
Baseline (after)	0.649767	0.602845	0.710984	0.889300	0.1526
FastText (before)	0.627099	0.517291	0.809132	0.886559	0.1925
FastText (after)	0.650314	0.611067	0.703391	0.886600	0.1508
Skip-Connection (before)	0.612131	0.493775	0.820423	0.885529	0.2062
Skip-Connection (after)	0.643874	0.599468	0.698885	0.885500	0.1541

A closer examination of macro precision and recall reveals that threshold optimization consistently improved macro precision across all models. In the baseline configuration, macro precision increased from 0.5216 to 0.6028. The FastText variant improved from 0.5173 to 0.6111, and the skip-connection model rose from 0.4938 to 0.5995. This substantial rise in precision indicates that the models became more selective after threshold adjustment, reducing false positive predictions.

As expected, macro recall decreased for all configurations: from 0.7987 to 0.7110 in the baseline model, from 0.8091 to 0.7034 in the FastText variant, and from 0.8204 to 0.6989 in the skip-connection model. This reflects the well-known precision–recall trade-off in multi-label classification, where increasing the decision threshold typically enhances precision while sacrificing some recall. Importantly, the overall improvement in macro F1 confirms that the net effect of threshold optimization was positive, yielding a more balanced and dependable classification outcome.

The AUC remained virtually unchanged before and after optimization. The baseline model maintained an AUC of 0.8893, while the FastText and skip-connection variants exhibited similar stability (0.8866 and 0.8855, respectively). This stability suggests that threshold tuning does not alter the models’ intrinsic ranking or discriminative capability; rather, it refines the operating point used for final predictions.

In contrast, Hamming Loss, which measures the proportion of incorrectly predicted labels, decreased substantially following threshold adjustment. For the baseline model, Hamming Loss declined from 0.1888 to 0.1526. The FastText variant improved from 0.1925 to 0.1508, and the skip-connection configuration decreased from 0.2062 to 0.1541. The consistent reduction in Hamming Loss indicates a meaningful decrease in overall label prediction errors, further demonstrating that threshold optimization plays a crucial role in enhancing the practical reliability of multi-label emotion classification models.

Taken together, these findings suggest that although all three architectures—baseline, FastText embedding, and skip-connection—demonstrated comparable performance after threshold optimization, the FastText embedding configuration achieved the strongest overall results across key metrics, including macro F1, macro precision, and Hamming Loss, marginally outperforming the baseline.

Importantly, the performance gap between the FastText and baseline configurations was minimal, indicating that both models are highly stable and effective for multi-label emotion

classification under the current experimental conditions. The improvements observed with FastText are incremental rather than transformative, but they consistently position it at the top across the primary evaluation criteria.

Accordingly, we recommend the FastText embedding configuration as the optimal approach for this task. Nevertheless, the baseline model remains a robust and practical alternative, particularly in scenarios where architectural simplicity, computational efficiency, or ease of deployment is prioritized.

5.9 Ensemble Methods

In addition to optimizing individual model architectures, we further investigated ensemble strategies to enhance robustness and predictive performance in multi-label emotion classification. A hard voting ensemble constructed from the top three model variants yielded consistent improvements over any single architecture.

Specifically, the ensemble achieved a macro F1 score of 0.6646, exceeding both the baseline and FastText configurations. Macro precision and macro recall increased to 0.6274 and 0.7122, respectively, indicating a well-balanced trade-off between precision and recall. The AUC also improved to 0.8988, reflecting strong discriminative capability across all emotion classes. Most notably, Hamming Loss decreased to 0.1435, representing the lowest overall misclassification rate among all tested configurations.

These findings demonstrate that the hard voting ensemble effectively leverages the complementary strengths of its constituent models, resulting in more stable and reliable predictions. Although the individual models already showed strong performance after threshold optimization, the ensemble consistently delivered measurable gains across key metrics.

Therefore, in practical scenarios where maximizing accuracy and robustness is critical, ensemble approaches such as hard voting constitute a well-justified and effective strategy for multi-label emotion classification.

6. Result and Comparison

The comprehensive comparison across all experiments reveals clear trends regarding the effectiveness of the approaches tested. Among individual models, the configuration combining KoELECTRA, Mecab tokenizer, FastText embeddings, and the reduced set of 37 emotion classes with optimized thresholding consistently delivered the best performance. This single-model setup achieved the highest macro F1, macro precision, and the lowest Hamming Loss, demonstrating robust and well-balanced multi-label classification capabilities.

Further improvements were realized through ensemble learning. A hard voting ensemble integrating predictions from the top three models outperformed all individual configurations across every primary evaluation metric. The ensemble achieved a macro F1 of 0.6646, macro precision of 0.6274, macro recall of 0.7122, AUC of 0.8988, and a Hamming Loss of 0.1435, representing the highest reliability and discriminative power observed throughout the study.

Comparing these results to the least effective baseline—KoELECTRA with default tokenizer, BCE loss, 43 emotions, and no optimization—highlights the magnitude of improvement. The baseline macro F1 was 0.5077, meaning the final ensemble represents an approximate 30.9% increase. Similarly, macro precision rose from 0.5264 to 0.6274 (+19.2%), macro recall from 0.4979 to 0.7122 (+43.1%), and AUC from 0.8253 to 0.8988 (+8.9%). Hamming Loss decreased slightly from 0.1450 to 0.1435, reflecting a modest but meaningful reduction in label prediction errors despite increased class complexity.

When compared to the previous KOTE benchmark, where KoELECTRA achieved a macro F1 of 0.56, macro precision of 0.54, macro recall of 0.61, AUC of 0.88, and MCC of 0.588, the ensemble developed in this study demonstrates clear superiority. The final ensemble improved macro F1 by ~18.7%, macro precision by 16.2%, macro recall by 16.7%, and AUC by 2.1%, while maintaining high reliability as measured by Hamming Loss and supporting indicators.

These results indicate that even in the challenging context of Korean multi-label emotion classification, substantial gains have been achieved in both class balance and classification precision, surpassing prior state-of-the-art benchmarks and confirming the effectiveness of the

7. Limitation

Despite the substantial advances presented in this study, several limitations must be acknowledged. First, the dataset composition introduces inherent biases and practical constraints. Over 80% of the final dataset is derived from KOTE, and the inclusion of external sources such as diaries or blog posts was limited by copyright restrictions, privacy concerns, and authors' reluctance to share personal writings publicly. These constraints make it difficult to ensure sufficient diversity in data sources, potentially causing the model to be sensitive to domain-specific patterns or label distributions, and limiting the generalizability of the findings to broader contexts in Korean text.

Second, although a nested 5-fold cross-validation approach was employed to mitigate data imbalance, it substantially increased computational time and experimental complexity. Due to constraints in time and computing resources, the exploration of alternative architectures, extensive data augmentation, or more exhaustive hyperparameter tuning was limited, leaving opportunities for further experimentation in future research.

Third, the criteria for class reduction and label merging were determined empirically by the research team, based on mean F1 scores and semantic similarity, rather than external linguistic validation or user studies. Consequently, there is potential for subjective bias in defining the final label taxonomy and merged classes.

Fourth, while ensemble methods were implemented, all constituent models were drawn from the same family of pre-trained architectures. The lack of structural diversity in the ensemble suggests that combining fundamentally different architectures or heterogeneous base models could potentially enhance robustness and predictive performance—an avenue not fully explored in this study.

Finally, all model evaluations were conducted on pre-segmented and pre-labeled datasets adhering to established annotation schemes. The impact of annotation noise, domain transfer effects, and deployment in real-world scenarios remains unexamined. Further studies are needed to fully assess the reliability and robustness of the models under unconstrained, real-world conditions.

8. Conclusion

In this study, we systematically advanced multi-label emotion classification for Korean diary entries and online comments. We enhanced model performance by integrating FastText embeddings and WBCE loss into a hybrid architecture based on KoELECTRA with the Mecab tokenizer. Through the application of 5-fold cross-validation, threshold optimization, and hard-voting ensembles, we increased the macro-F1 score from 0.56 to 0.66 relative to the original KOTE benchmark. In addition, we consolidated labels from 43 to 37 classes and augmented the training data with both synthetic and real diary entries, improving prediction stability, particularly for rare emotions. This comprehensive approach lays the groundwork for a nuanced emotion recognition system tailored to Korean linguistic and cultural contexts, and paves the way for future research on heterogeneous model ensembles, domain adaptation, and real-time emotion tracking.

Funding Statement This research was supported by Fund of Biomedical Research Institute, Jeonbuk National University Hospital.

References

- Acheampong F. A., Chen W. and Nunoo-Mensah, H. (2020). Text-based emotion detection: Advances, challenges, and opportunities. *Engineering Reports* 2(7), e12189.
- Akar S., Kim Y. S. and Noh M. J. (2024). Sentiment analysis on 'HelloTalk' app reviews using NRC Emotion Lexicon and GoEmotions dataset. *Smart Media Journal* 13(6), 35–43.
- Ameer I., Bölücü N., Siddiqui M. H. F., Can B., Sidorov G. and Gelbukh A. (2023). Multi-label emotion classification in texts using transfer learning. *Expert Systems with Applications* 202, 118534.
- Arlot S. and Celisse A. (2010). A survey of cross-validation procedures for model selection. *Statistics Surveys* 4, 40–79.
- Bakker A. B. and Daniels K. (2013). Introduction: A day in the life of a happy worker. In A. B. Bakker and K. Daniels

- (Eds.), *A day in the life of a happy worker* (pp. 1–7). Psychology Press.
- Ben-Baruch E., Ridnik T., Zamir N., Noy A., Friedman I., Protter M. and Zelnik-Manor L.** (2021). Asymmetric Loss for Multi-Label Classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 82–91.
- Bolger N., Davis A. and Rafaeli E.** (2003). Diary methods: Capturing life as it is lived. *Annual Review of Psychology* **54**(1), 579–616.
- Briner R. B. and Kiefer T.** (2005). Psychological research into the experience of emotion at work: Definitely older, but are we any wiser? In N. M. Ashkanasy, W. J. Zerbe and C. E. J. Härtel (Eds.), *The effect of affect in organizational settings* (Vol. 1, pp. 281–307). Emerald Group Publishing.
- Chae S.** (2019). *A Study on Text Embedding Methodology for Korean Sentiment Analysis* (Master's thesis). University of Seoul, Seoul, Korea.
- Clark K., Luong M.-T., Le Q. V. and Manning C. D.** (2020). ELECTRA: Pre-training text encoders as discriminators rather than generators. In *Proceedings of the 8th International Conference on Learning Representations (ICLR 2020)*.
- Cowen A. S., and Keltner D.** (2017). Self-report captures 27 distinct categories of emotion bridged by continuous gradients. In *Proceedings of the National Academy of Sciences of the United States of America* **114**(38), E7900–E7909.
- Demszky D., Movshovitz-Attias D., Ko J., Cowen A., Nemade G. and Ravi S.** (2020). GoEmotions: A dataset of fine-grained emotions. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4040–4054.
- Devlin J., Chang M.-W., Lee K. and Toutanova K.** (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In J. Burstein, C. Doran and T. Solorio (Eds.), *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (Vol. 1, pp. 4171–4186). Association for Computational Linguistics.
- Gendron M., Roberson D., van der Vyver J. M. and Barrett L. F.** (2014). Perceptions of emotion from facial expressions are not culturally universal: Evidence from a remote culture. *Emotion* **14**(2), 251–262.
- Grandvalet Y. and Bengio Y.** (2004). Semi-supervised learning by entropy minimization. In *Advances in Neural Information Processing Systems (NIPS 2004)*, vol. 17, pp. 529–536.
- Han B., Li Z., Wang X., Hou Y., Liu H., Hou J. and Jia Y.** (2025). DiCaP: Distribution-calibrated pseudo-labeling for semi-supervised multi-label learning. *arXiv*, arXiv:2511.20225.
- He K., Zhang X., Ren S. and Sun J.** (2015). Deep Residual Learning for Image Recognition. *arXiv*, arXiv:1512.03385.
- Hinton G. E. and Salakhutdinov R. R.** (2006). Reducing the Dimensionality of Data with Neural Networks. *Science* **313**(5786), 504–507.
- Izard C. E., Libero D. Z., Putnam P. and Haynes O. M.** (1993). Stability of emotion experiences and their relations to traits of personality. *Journal of Personality and Social Psychology* **64**(5), 847–860.
- Jabreel M. and Moreno A.** (2019). A deep learning-based approach for multi-label emotion classification in Tweets. *Applied Sciences* **9**(6), 1123.
- Jang J. M., Min J. O. and Noh H. S.** (2022). KorPatELECTRA: A pre-trained language model for Korean patent literature to improve performance in the field of natural language processing (Korean Patent ELECTRA). *Journal of the Korea Society of Computer and Information* **27**(2), 15–23.
- Jeon D., Lee J. and Kim C.** (2022). User Guide for KOTE: Korean Online Comments Emotions Dataset. Department of Psychology, Seoul National University.
- Jeon T.** (2022). A linguistic study on Korean text tokenization methods: Using fastText word embedding. *Language Facts and Perspectives* **55**, 309–354.
- Kim M. and Kim N.** (2021). Label embedding for improving classification accuracy using AutoEncoder with skip-connections. *Journal of Intelligence and Information Systems* **27**(3), 175–197.
- Kim H. S., Han H. J. and Kim H. S.** (2024). The structure of emotion regulation strategies among Korean adults. *Personality and Individual Differences* **229**, 112752.
- Lee D.-H.** (2013). Pseudo-label: The humble adversarial training for semi-supervised learning. In *ICML 2013 Workshop on Challenges in Representation Learning*.
- Lee J.** (2020). KcBERT: Korean comments BERT. In *Proceedings of the 32nd Annual Conference on Human and Cognitive Language Technology*, pp. 437–440.
- Lee M.** (2023). Psychological effects of MBTI uses among undergraduate students: The impacts of MBTI trust on self-determination motivation, self-perception, interpersonal communication competency, and subjective well-being. *Korean Journal of Communication and Information* **122**, 133–160.
- Lee S., Jang H., Baik Y., Park S. and Shin H.** (2020). KR-BERT: A small-scale Korean-specific language model. *arXiv*, arXiv:2008.03979.
- Martínez-Tejada L. A., Peña-Castillo M. and Sanchez J.** (2020). Analysis of personality and EEG features in emotion recognition using machine learning techniques to classify arousal and valence labels. *Sensors* **20**(6), 1603.
- Malik U., Bernard S., Pauchet A., Chatelain C., Picot-Clément R. and Cortinovis J.** (2024). Pseudo-labeling with large language models for multi-label emotion classification of French tweets. *IEEE Access* **12**, 15902–15916.
- Mesquita B. and Frijda N. H.** (1992). Cultural variations in emotions: A review. *Psychological Bulletin* **112**(2), 179–204.
- Mousavi S. S., Ghassem-Sani G., Saeidi M. and Ruiz I. V. M.** (2023). Pseudo-labeling with large language models for multi-label emotion classification of French Tweets. *IEEE Access* **11**, 60579–60589.
- Nadin S. and Cassell C.** (2006). The use of a research diary as a tool for reflexive practice: Some reflections from management research. *Qualitative Research in Accounting Management* **3**(3), 208–217.

- Ortony A., Clore G. L. and Collins A.** (1990). The cognitive structure of emotions. Cambridge University Press.
- Pang B., Lee L. and Vaithyanathan S.** (2002). Thumbs up? Sentiment classification using machine learning techniques. In *Proceedings of the ACL-02 Conference on Empirical Methods in Natural Language Processing*, pp. 79–86.
- Park E. L. and Cho S.** (2014). KoNLPy: Korean natural language processing in Python. In *Proceedings of the 26th Annual Conference on Human Cognitive Language Technology*.
- Park J.** (2020). KoELECTRA: Pretrained ELECTRA model for Korean. GitHub Repository. Available online: <https://github.com/monologg/KoELECTRA>.
- Park S., Lee S., Kim W. and Song M.** (2022). A deep learning-based depression trend analysis of Korean on social media. *Journal of the Korean Society for Information Management* **39**(1), 91–117.
- Patki N., Wedge R. and Veeramachaneni K.** (2016). The Synthetic Data Vault. MIT CSAIL.
- Revelle W. and Scherer K. R.** (2009). Personality and emotion. In D. Sander and K. R. Scherer (Eds.), *The Oxford Companion to Emotion and the Affective Sciences* (pp. 304–306). Oxford University Press.
- Rezaei-Dastjerdehi M., Mijani A. and Fatemizadeh E.** (2020). Addressing imbalance in multi-label classification using weighted cross entropy loss function. In *Proceedings of the IEEE International Conference on Biomedical and Health Informatics (ICBME)*, pp. 333–338.
- Rizve M. N., Duarte K., Rawat Y. S. and Shah M.** (2021). In defense of pseudo-labeling: An uncertainty-aware pseudo-label selection framework for semi-supervised learning. In *International Conference on Learning Representations (ICLR 2021)*.
- Scherer K. R., Wranik T., Sangsue J., Tran V. and Scherer U.** (2004). Emotions in everyday life: Probability of occurrence, risk factors, appraisal and reaction patterns. *Social Science Information* **43**(4), 499–570.
- Shim K. and Yang J.** (2002). Mach: A supersonic Korean morphological analyzer. In *Proceedings of COLING 2002: The 19th International Conference on Computational Linguistics*.
- SKTBrain.** (2019). KoBERT. GitHub Repository. Available online: <https://github.com/SKTBrain/KoBERT>.
- Soppari K., Hussain S., Surya G. V. N. and Rao M. P.** (2025). EMO diary: daily diary with sentiment analysis. *International Journal of Scientific Research Engineering Trends* **11**(1), 304–317.
- Strapparava C. and Mihalcea R.** (2007). SemEval-2007 task 14: Affective text. In *Proceedings of the Fourth International Workshop on Semantic Evaluations (SemEval-2007)*, pp. 70–74.
- Turney P. D.** (2002). Thumbs up or thumbs down? Semantic orientation applied to unsupervised classification of reviews. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pp. 417–424.
- Ubani S., Polat S. O. and Nielsen R. D.** (2023). ZeroShotDataAug: Generating and augmenting training data with ChatGPT. *arXiv*, arXiv:2304.14334.
- Wu T., Huang Q., Liu Z., Wang Y. and Lin D.** (2020). Distribution-balanced loss for multi-label classification in long-tailed datasets. In *European Conference on Computer Vision (ECCV 2020)*, Lecture Notes in Computer Science, vol. 12363, pp. 162–178.
- Yang K.** (2021). Transformer-based Korean pretrained language models: A survey on three years of progress. *arXiv*, arXiv:2112.03014
- Yang L., Huang B., Guo S., Lin Y. and Zhao T.** (2023). A Small-sample text classification model based on pseudo-label fusion clustering algorithm. *Applied Sciences* **13**(8), 4716.
- Yoon S. and Kim B.** (2025). Adaptive intervention architecture for psychological manipulation detection: A culture-specific approach for adolescent digital communications. *Information* **16**(5), 379.
- Yu K., Lin T. R., Ma H., Li X. and Li X.** (2021). A multi-stage semi-supervised learning approach for intelligent fault diagnosis of rolling bearing using data augmentation and metric learning. *Mechanical Systems and Signal Processing* **146**, 107043.